

双通道 10 位 DA 波形发生器

1、实验目的：

双通道 10bitDA。

2、实验原理：

程序顶层原理图文件如下 1

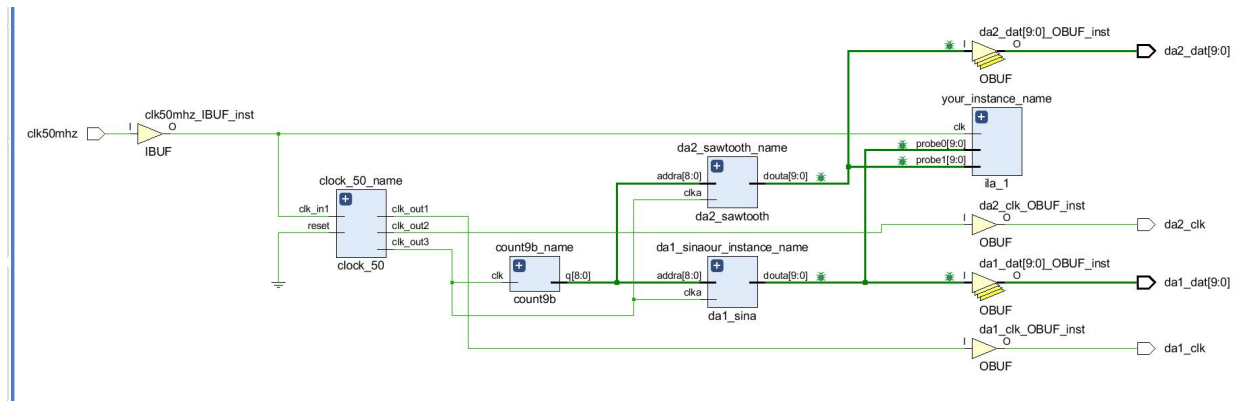


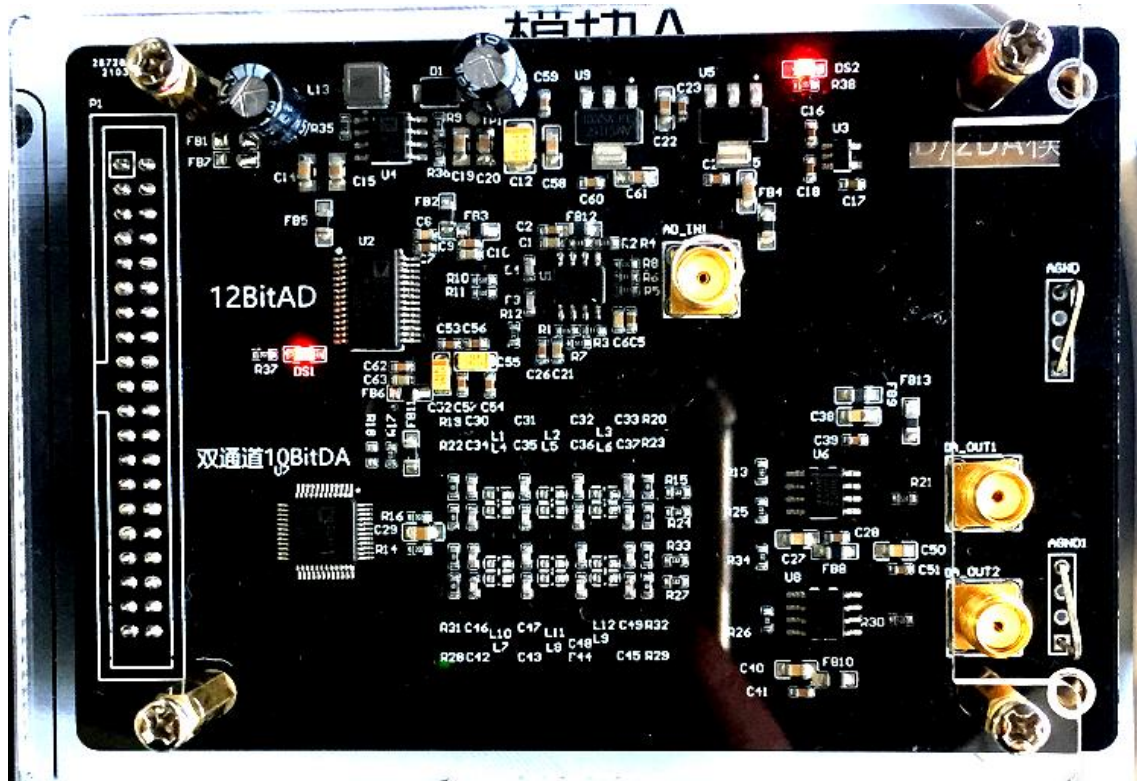
图 1 RTL 原理图

1. da 逻辑分析仪波形图 2:



图 2

3、硬件准备：



4

4、管脚约束:

```

set_pin_assignment { clk50mhz } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE = PULLUP; }
set_pin_assignment { da1_clk } { LOCATION = H15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;
PULLTYPE = NONE; }
set_pin_assignment { da1_dat[0] } { LOCATION = H16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[1] } { LOCATION = K15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[2] } { LOCATION = J16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[3] } { LOCATION = G16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[4] } { LOCATION = G14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[5] } { LOCATION = F14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[6] } { LOCATION = F15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[7] } { LOCATION = D16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da1_dat[8] } { LOCATION = E15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
  
```

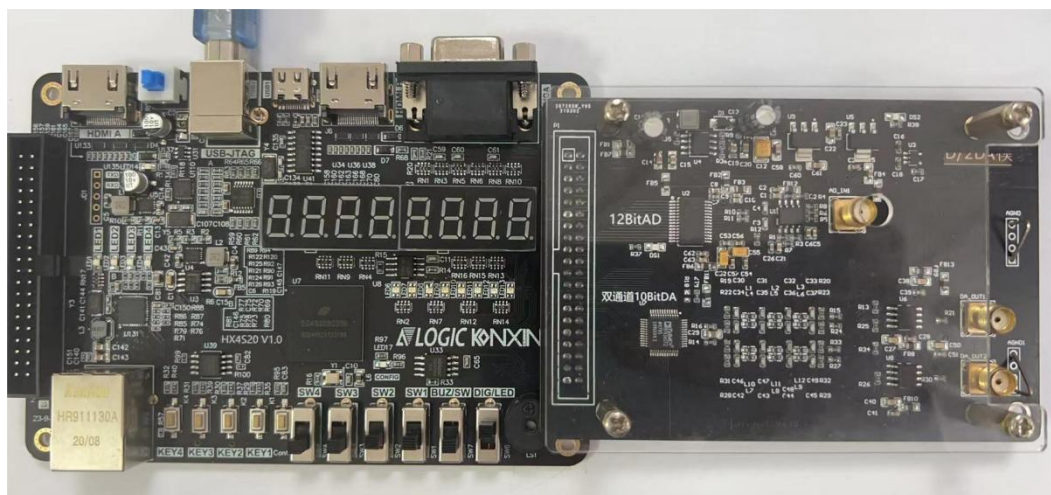
```

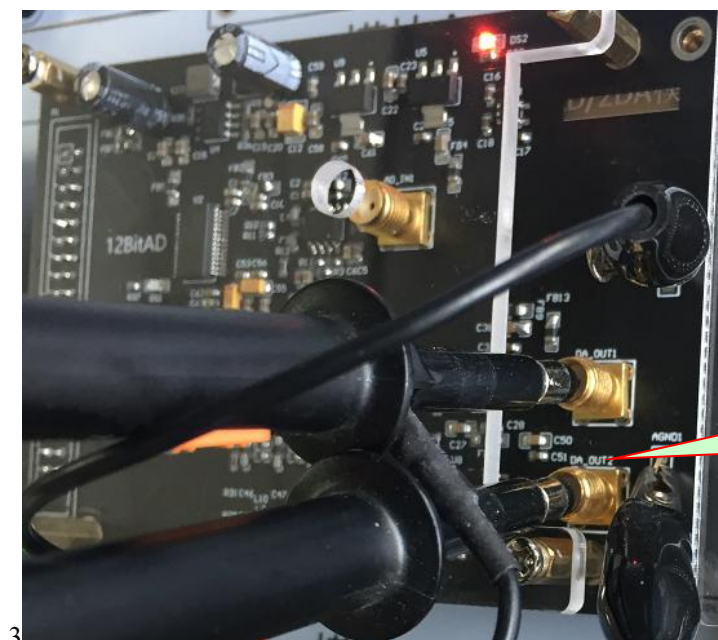
set_pin_assignment { da1_dat[9] } { LOCATION = L10; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_clk } { LOCATION = M16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH = 8;
PULLTYPE = NONE; }
set_pin_assignment { da2_dat[0] } { LOCATION = T15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[1] } { LOCATION = R16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[2] } { LOCATION = R15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[3] } { LOCATION = P16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[4] } { LOCATION = P15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[5] } { LOCATION = N16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[6] } { LOCATION = N14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[7] } { LOCATION = M15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[8] } { LOCATION = L16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { da2_dat[9] } { LOCATION = K16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }

```

5、硬件连接

实验的硬件连接图 3：扩展板接插在 J1 口上，示波器正级连接 2 个 SMA 端



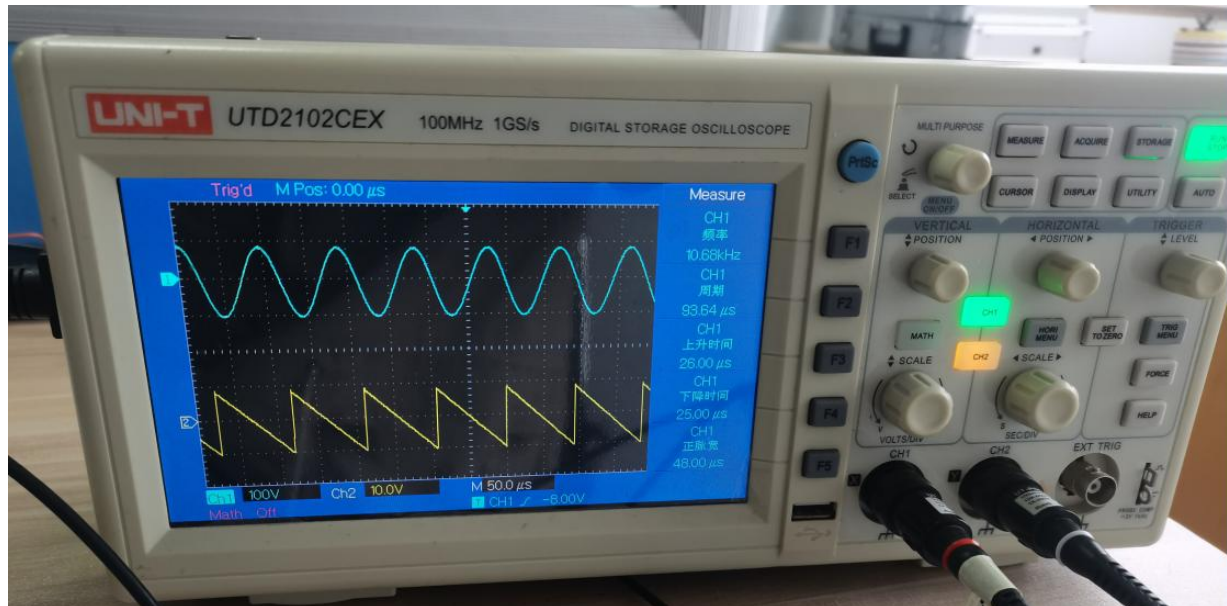


DA 输出, 右边示波器正极(钩)接正极, 黑夹示波器负极接地

图 3

6、实验结果:

巨齿波信号。



26-6